Heat-Aware Pedestrian Routing with Raster-Derived Exposure and Hard Detour Budgets: A Washington, DC Pilot

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Abstract

Heat-aware walking routes trade modeled thermal exposure against extra distance. This pilot compares four environmental objective settings with shortest distance on a Washington, DC walking network, using Universal Thermal Climate Index (UTCI) and shadow rasters at one modeled hour. Five methods, 500 origin–destination pairs, and four distance allowances produce 10,000 routing attempts. On the 2,000 cases shared by all methods, mean reductions in length-weighted UTCI exceedance above 26°C are 0.92–0.96 K, with 4.84–6.55% extra distance. We evaluate both average exceedance and its length-scaled total, report capped-search fallback separately, and test weight, threshold, and sampling changes on a fixed 100-pair subset. Changing 20 m to 5 m sampling changes 78 of 100 paths. Exact-reference tests quantify losses from capped search on deliberately difficult generated graphs. Code, frozen input checksums, and route records are shared. The contribution is a reproducible comparison of objective choices within one solver. The study does not establish transfer beyond this snapshot or validate physiological benefit.

Keywords

pedestrian routing, UTCI, shade, detour constraints, computational benchmark

1 Introduction

Two walking routes of similar length can have different shade and thermal conditions. A route that avoids exposed streets may reduce modeled heat exposure, but a long detour also adds walking. An explicit distance allowance makes this tradeoff understandable: a traveler can ask for a cooler route no more than 10% longer than the shortest path.

This paper examines that formulation in a Washington, DC pilot. It attaches modeled UTCI and direct-shadow rasters to a pedestrian graph, then compares objectives that emphasize UTCI, shade, or a composite of environmental features. We study how these objective settings behave within one capped search implementation, using established hard-detour constraints and cost–distance labels.

The experiment uses 500 ordered origin–destination (OD) pairs, four detour allowances, and five methods. We compare routes on the same OD–budget cases and distinguish optimized routes from shortest-path fallback. We also test coefficient choices and sampling on a fixed subset, and check search outcomes against exact references on generated graphs.

The benchmark asks how much the selected objective changes modeled exceedance, added distance, and shade under a hard distance ceiling. It reports both average exceedance and an exceedance–distance index, because a cooler average can accompany more total walking through heat. The numerical results use a corrected rerun; the earlier archive is retained separately for audit.

2 Related Work

Shade-aware routing predates this study. Vartholomaios derives least-cost paths from SOLWEIG shadows [24]; Da et al. use image-derived shade [5]; and Kolaxidis et al. incorporate solar exposure into pedestrian routing [16]. CoolWalks studies sunlit-distance penalties and the difference between centerline and detailed pedestrian networks [26]. Yang et al. incorporate time-varying building shadows [27], whereas this pilot holds the environmental field fixed.

Other work uses thermal indices. Colaninno et al. combine sidewalk mobility and microclimate modeling [4]. Patel et al. rank Google Maps candidate routes using SOLWEIG-GPU UTCI [22]. CoolPaths minimizes summed edge length times Physiological Equivalent Temperature (PET) under a detour limit [25]. Here, the primary reported metric is length-normalized UTCI exceedance. The routing formulation follows established resource-constrained shortest-path methods [14]. More generally, Skoumas et al. enrich route computation with contextual information about places [23]; environmental attributes are another application of that idea.

UTCI converts meteorological and radiative conditions into an equivalent temperature under standardized reference assumptions [3]. SOLWEIG models urban radiation and mean radiant temperature [18]. Evidence that heat influences route choice [1] and that pedestrians seek shade [17] motivates these routing preferences. It does not calibrate this paper's weights or validate its route-level outcomes for an individual pedestrian.

3 Routing Pipeline

3.1 Graph and distance budget

Let G_0 be the original directed walking multigraph. For a route P, let $L(P) = \sum_{e \in P} l_e$, where l_e is edge length in metres. For an OD pair, define the original-network shortest distance L_0^* and the final ceiling

$$L_0^* = \min_{P \subset G_0} L(P), \qquad D_0 = (1 + \delta) L_0^*. \tag{1}$$

The minimization is over directed paths connecting that pair. The allowance $\delta \in \{0.05, 0.10, 0.20, 0.30\}$ limits proportional extra distance.

To reduce feature-extraction work, a node v is retained when

$$d_{G_0}(o, v) + d_{G_0}(v, d) \le D_0 + 50\,\text{m}, \tag{2}$$

where d_G denotes shortest-path distance on graph G. The pipeline then assigns raster-backed features and removes edges lacking valid UTCI samples. The resulting feature graph is G_f. Network processing uses OpenStreetMap, OSMnx, and NetworkX [2, 13, 21].

3.2 Sampling and reported exposure

Each source segment in an edge geometry is subdivided to a maximum spacing of 20 m, retaining endpoints. Coordinates are transformed into each raster's coordinate reference system before sampling. Non-finite and NoData samples are omitted. Edges without UTCI samples are removed; edges without direct-shadow samples use the surrogate in Appendix B.

For UTCI samples u_{ei} on edge e, define

$$\bar{u}_e = \frac{1}{n_e^u} \sum_i u_{ei}, \qquad u_e^{\max} = \max_i u_{ei}, \tag{3}$$

$$h_e = \frac{1}{n_e^u} \sum_i \max(0, u_{ei} - \tau), \quad \tau = 26^\circ\mathrm{C}. \tag{4}$$

For valid shadow samples b_{ej}, the direct-shade fraction is $s_e^{\mathrm{dir}} = (n_e^b)^{-1} \sum_j \mathbf{1}[b_{ej} \le 0.5]$; the raster convention is zero for shade and one for sun. Except for hybrid azimuth, each method uses this value when present and the surrogate s_e^{az} otherwise. The hybrid-azimuth method uses s_e^{az} everywhere.

The 26°C reference is the conventional UTCI boundary between no thermal stress and moderate heat stress [12]. It supplies an engineering reference rather than a personalized safety threshold. The main run uses a 20 m maximum interval on nominal 1 m rasters; Section 4.4 compares alternative intervals against 5 m sampling on the same selected paths.

Reported route quantities are length-weighted edge summaries:

$$H(P) = \frac{\sum_e l_e h_e}{L(P)}, \qquad \bar{U}(P) = \frac{\sum_e l_e \bar{u}_e}{L(P)}, \tag{5}$$

$$S(P) = \frac{\sum_e l_e s_e}{L(P)}, \qquad U^{\max}(P) = \max_e u_e^{\max}. \tag{6}$$

H is mean positive UTCI exceedance, expressed in kelvins; $\bar{U}$ and $U^{\max}$ are UTCI values in degrees Celsius. S is a shade fraction. Because sampling restarts at segment boundaries, these quantities are numerical proxies rather than uniform line integrals. As a post hoc check of normalization, we also calculate $J(P) = L(P)H(P)$, an exceedance–distance index in K m. It tests whether a lower route average offsets added distance. Neither H nor J is a validated physiological dose.

3.3 Objectives and method variants

We define the composite cost as an engineering score. It combines distance and time with thermal, scenic, traffic, and slope penalties; Appendix C specifies the terms and normalizations. All are nonnegative. The thermal term is

$$\phi_e = \frac{[\bar{u}_e - \tau]_+}{8\,\mathrm{K}} + \frac{[u_e^{\max} - \tau]_+}{10\,\mathrm{K}} + \frac{h_e}{6\,\mathrm{K}} + 0.45(1 - s_e), \tag{7}$$

where $[x]_+ = \max(0, x)$. Its correlated UTCI and shade components can count related radiative information more than once. Neither its coefficients nor the contextual proxies are fitted to route-choice or physiological observations.

Table 1: Configured objective variants. A_e is the distance–time term in Appendix C; all environmental penalties also use its edge-length scale.

Method	Selection objective
Shortest	Dijkstra search minimizing $L(P)$.
Hybrid azimuth	A_e plus twice the thermal term, using s_e^{az} on all edges.
Direct shadow	A_e plus twice the shade-only part of ϕ_e.
UTCI exceedance	A_e plus twice the $h_e/(6\,\mathrm{K})$ part of ϕ_e.
Composite	Full thermal, scenic, traffic, and slope cost.

Table 1 defines the four objective variants of this implementation. Distance and time remain active in all four. The focused methods are therefore not pure single-feature optimizations. For reporting, selected routes are evaluated with full features; the hybrid shade outcome remains a surrogate and is not comparable with direct shade.

3.4 Capped search and fallback

The solver first computes an unconstrained minimum-cost path and returns it if it meets D_0. Otherwise it explores cost–distance labels in increasing cost order. At a node, a label dominates another if it has no greater length and no greater cost. Labels are pruned when accumulated length plus shortest remaining distance exceeds D_0. Paths may not repeat nodes.

The dominance registry retains at most eight labels per node, ordered by cost then length. A newly queued label can survive even when omitted from that registry, so this does not bound the queue size. The label loop stops after 20,000 queue removals or 1.5 s, with timing beginning after the initial cost-path calculation. A capped search returns a feasible length-shortest fallback when available. If feature removal leaves no path within the original ceiling, the request fails. The final acceptance check enforces $L(P) \le D_0$ within 10^{-6} m numerical tolerance.

We record whether a returned route came from the unconstrained cost path, label search, or shortest-path fallback, along with cap termination and failure type. The caps preclude a general optimality guarantee and do not bound end-to-end latency. Appendix A compares the search with exact enumeration on small graphs.

Before this rerun, the implementation was corrected to propagate the original-network ceiling, choose shortest parallel edges for shortest-path fallback, preserve sampled exceedance through objective projection, and use above-ground building heights. In particular, $\mathbb{E}[\max(0, U - \tau)]$ cannot be replaced with $\max(0, \mathbb{E}[U] - \tau)$. The earlier run lacked complete execution provenance; its records remain separate from the results below.

Table 2: Input layers. Different source years and street-centerline sampling limit spatial interpretation.

Layer	Source and role
Walking network	OpenStreetMap via Geofabrik/OSMnx; directed centerlines, not verified sidewalks [2, 21].
UTCI and shadow	SOLWEIG-GPU modeled snapshot; thermal and direct-shade features [15].
Terrain/buildings	NOAA-hosted DC OCTO 2022 LiDAR and DEM; grade and height proxies [19, 20].
Canopy/land cover	LiDAR canopy heights and DC 2020 layers; greenery, water, and surrogate shade [6, 7].
Traffic	DDOT 2023 AADT lines; normalized traffic proxy, not sound measurements [9].

4 Data and Analysis

4.1 Study domain and inputs

The study uses one modeled Washington, DC snapshot labeled June 19, 2026, 14:00. The rerun explicitly uses UTC−04 : 00 for the hybrid solar calculation. Sustainable DC 2.0 reports a 13.3% walking commute share in 2016 [11]; the sampled OD pairs nevertheless represent covered network nodes, not observed trips.

Table 2 lists the input sources. SOLWEIG-GPU [15] generated the nominal 1 m UTCI and shadow fields from urban-surface inputs and hourly Open-Meteo forcing [28]. These fields are held fixed and have no independent validation here. We recovered the versioned graph and rasters through Git LFS, recorded their SHA-256 checksums, and reproduced all 500 archived shortest distances exactly. The original graph has 61,543 nodes and 174,718 directed edges. For the hybrid proxy, building height is $\max(0, \mathrm{DSM}-\mathrm{DEM})$ on the aligned source grids; absolute surface elevation must not be interpreted as height above ground. Frozen routing inputs enable rerouting, but the retained provenance still lacks the forecast retrieval time, selected forecast model, and environmental-model release needed to regenerate the thermal field exactly.

4.2 OD sample and experimental design

The original generator uses the largest weakly connected graph component, then retains endpoints with finite UTCI and shadow raster values. With seed 42, it samples 500 distinct ordered node pairs whose shortest network distances lie between 400 and 3,000 m. We reuse those exact pairs. Their shortest distances range from 412.9 to 2,998.9 m. Figure 1 locates the endpoints and shows the network-distance distribution used for eligibility. Large parts of the District have no sampled endpoints, so these locations do not establish district-wide coverage. Both maps use EPSG:32618 and frozen DDOT/DC GIS roads and water for geographic context; those display layers do not enter routing or exposure calculations [8, 10].

Each pair is evaluated with five methods at four allowances, giving 10,000 attempts. The 2,000 shortest-distance records repeat 500 pairs; the analysis checks their edge sequences across allowances. Environmental routes can change with allowance-specific corridors, features, and search. Records from the same OD pair are not independent observations. Eight deterministic shards assign pair index i to i mod 8, retaining every method and allowance for that pair. Each shard records its configuration, source-file hashes, and software versions. Timing includes corridor construction, feature extraction, and routing; direct-shadow methods share a persistent feature cache, while hybrid azimuth uses a separate cache.

4.3 Matched comparison and uncertainty

Let $P^0_{i\delta}$ be the matched shortest route and $P^m_{i\delta}$ the route returned by method m. Positive improvements are

$$\Delta H_{i\delta m} = H(P^0_{i\delta}) - H(P^m_{i\delta}), \tag{8}$$

with analogous $\Delta\bar{U}$; shade change reverses the order. Extra distance is $100(L(P^m_{i\delta})/L^*_0 - 1)$.

At each allowance, the primary comparison uses the intersection $\mathcal{C}_\delta$ of successful records for all five methods. Matching includes study, configuration hash, run identifier, raster timestamp, OD identifier, and allowance; duplicates are rejected. Shards retain their original identifiers, and matching stays within each shard. The common set contains 2,000 of 2,000 pair–budget cases from 500 OD pairs. Pooled means weight these cases equally and include shortest-path fallbacks. We also count records with lower H but higher J, using 10^{-9} K and 10^{-6} K m tolerances to exclude rounding noise.

For a paired method contrast, we compute the mean difference in recorded H on the same common records. Percentile intervals use 2,000 bootstrap resamples of whole OD pairs, retaining each pair's available allowances, with seed 42. These exploratory 95% intervals are not adjusted for multiple comparisons. They summarize variability within this one modeled snapshot and do not account for microclimate error or selection of the successful subset.

4.4 Sensitivity design

We use the first 100 serialized OD pairs at a 20% allowance for 16 settings: the reference composite; removal and doubling of each of its thermal, shade, scenic, traffic, and slope coefficients; thresholds of 24 and 28°C; and maximum sample intervals of 5, 10, and 40 m. Each coefficient changes alone. Distance and time weights remain fixed. These exploratory choices test specific dependencies without fitting a preferred setting to the results.

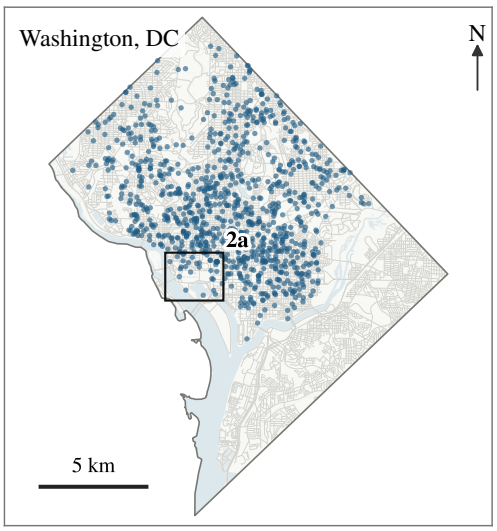
Every selected path is rescored on the same unperturbed field with $\tau = 26$°C, first at 20 m and then at 5 m. Path changes use the ordered edge sequence. For a sampling setting, we compare its own H with the 5 m score of that same path, separating metric disagreement from rerouting. The 5 m calculation is a denser numerical comparator, not ground truth or a proof of convergence at 1 m. Failures remain in the denominator, and dense-rescoring coverage is checked separately. The repeated reference setting also checks whether rerouting under the caps changes its own control path.

5 Results

5.1 Feasibility and route sources

The corrected run returns 10,000 routes from 10,000 attempts, all within their original-network distance ceilings. Figure 3a separates the initial cost path, label-search result, and shortest-path fallback. Environmental fallback accounts for 5.55–14.90% of attempts across the four settings. These are feasible walking routes;

(a) Sampled endpoints



Blue: sampled endpoints; box: Fig. 2a map extent

(b) Network-distance eligibility

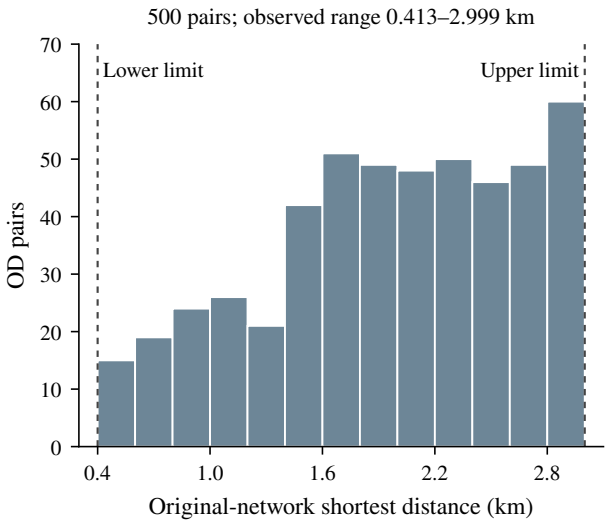


Figure 1: Sample and distance eligibility. (a) Endpoint records for 500 OD pairs within the District; repeated endpoints can overlap. The box locates Figure 2a. (b) Original-network shortest distances in 200 m bins, bounded by the 400–3,000 m sampling limits. Distances are network lengths, not straight-line separation.

fallback does not establish that a cooler route was found. The common comparison includes every such fallback.

The archived run accepted 6,093 of 8,000 environmental attempts and logged an expansion-limit error for each rejection. The corrected rerun records cap termination and returns a feasible fallback instead of treating that event as geographic infeasibility. Several implementation changes and an explicit solar-time offset separate the runs, so this comparison does not isolate the effect of any single fix.

5.2 Exposure and added distance

Table 3 compares the same 2,000 OD–budget cases for every method. Mean exceedance reductions range from 0.92–0.96 K, with 4.84–6.55% extra distance. The shortest-route mean is 4.794 K. The composite has 0.009 K lower mean H than the UTCI-focused setting (OD-cluster interval [−0.015, 0.035] K). The interval includes zero. The UTCI-focused setting has 0.028 K lower mean H than the direct-shadow setting (interval [0.010, 0.045] K).

Figure 2 links one recorded route comparison with all 500 common pairs at a 20% allowance. The mapped example adds 13.1% distance and lowers H by 1.06 K. None has higher H than its matched shortest route. Selection minimizes a sum of distance, time, and length-scaled thermal costs; it does not directly minimize the normalized H.

Lower mean exceedance can accompany a higher exceedance–distance index. Direct shadow produces 26 such records (1.3%). The composite has one such record; we find none for the UTCI setting. Table 3 reports both H and J. Hybrid shade uses a surrogate, so its ΔS is omitted; the raw records also report how much direct-shadow routing relies on surrogate fallback.

5.3 Distance allowances

For the composite, mean ΔH changes from 0.705 K at a 5% allowance to 1.137 K at 30%, with mean extra distance of 2.43% and 6.63%, respectively. These summaries use the same matched OD pairs at each allowance. A larger ceiling expands the feasible path set on a fixed graph. The implementation also rebuilds its corridor and applies caps, so the returned route need not improve monotonically for an individual pair.

5.4 Weights, threshold, and sampling

All 1,600 sensitivity attempts succeed, and every selected path has 5 m rescoring. Table 4 shows the one-factor weight changes. Removing or doubling a coefficient changes 13–95 of 100 common paths, depending on the coefficient. All exposure contrasts in that table use $\tau = 26^{\circ}$C and 5 m samples, so the numbers reflect selected paths rather than changed reporting definitions.

Changing the selection threshold to 24 and 28°C changes 44 and 32 paths, respectively. On the common 5 m reporting field, mean H is 0.098 K higher than the default at 24°C and 0.024 K lower at 28°C. These results quantify sensitivity to two nearby references; they do not calibrate a personal heat-stress threshold.

Table 5 compares sampling intervals on the same selected paths. At 20 m, the median absolute difference from 5 m is 0.099 K and the 90th percentile is 0.252 K. Switching to 5 m changes 78 of 100 paths,

Table 3: Matched outcomes on the same 2,000 OD–budget cases. Positive ΔH and ΔJ favor the environmental route; ΔS is added shade. Intervals resample whole OD pairs. Hybrid shade uses a different surrogate, so its comparison is omitted. Fallbacks are included.

Method	Mean ΔH (K) [95% interval]	Mean ΔJ (K km)	Mean ΔS	Extra distance (%)
Shortest	0.000	0.000	0.000	0.00
Hybrid azimuth	0.962 [0.911, 1.016]	1.518	—	5.61
Direct shadow	0.922 [0.874, 0.974]	1.352	0.143	6.55
UTCI exceedance	0.950 [0.900, 1.004]	1.421	0.139	6.35
Composite	0.959 [0.907, 1.013]	1.572	0.139	4.84

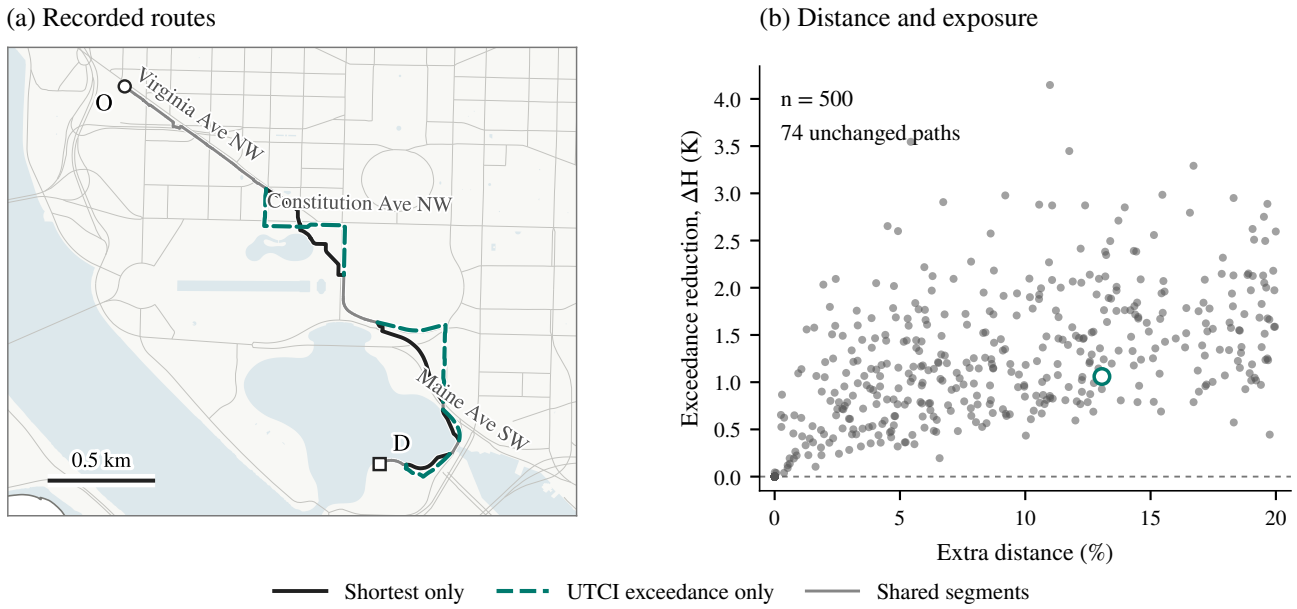


Figure 2: Route change and paired outcomes at a 20% distance allowance. (a) Recorded paths for `od_00000`, selected by the median paired exceedance reduction. O/D denote origin/destination; gray segments are shared. (b) Every point is a common OD pair. Positive $\Delta H = H_{\text{shortest}} - H_{\text{UTCI}}$ means lower mean positive exceedance above 26°C. The outlined point identifies (a). Map coordinates use EPSG:32618.

Table 4: One-factor composite sensitivity on 100 complete paired paths, scored at 5 m and 26°C. Entries show removed/doubled coefficients. Positive ΔH means lower exceedance than the default.

Coefficient	Changed paths	ΔH (K)
Thermal	95/39	−0.580/0.050
Shade	18/17	−0.011/0.016
Scenic	43/36	−0.010/ − 0.008
Traffic	29/13	0.038/ − 0.018
Slope	19/13	−0.001/ − 0.028

so the 20 m choice is not stable at the route level. The repeated reference returns identical paths.

Table 5: Sampling disagreement with 5 m scoring of each same path, on 100 complete cases. Changes compare edge sequences with the 20 m default. The 5 m error is zero by construction.

Spacing (m)	Median error (K)	90th pct. (K)	Changed
5	0.000	0.000	78
10	0.057	0.132	78
20	0.099	0.252	0
40	0.168	0.410	73

5.5 Observed runtime

Figure 3b shows every attempt, including every fallback. Method medians range from 0.180–0.729 s; 90th percentiles range from 0.702–2.459 s. The complete observed maximum is 7.564 s. These are end-to-end measurements from the rerun, including corridor

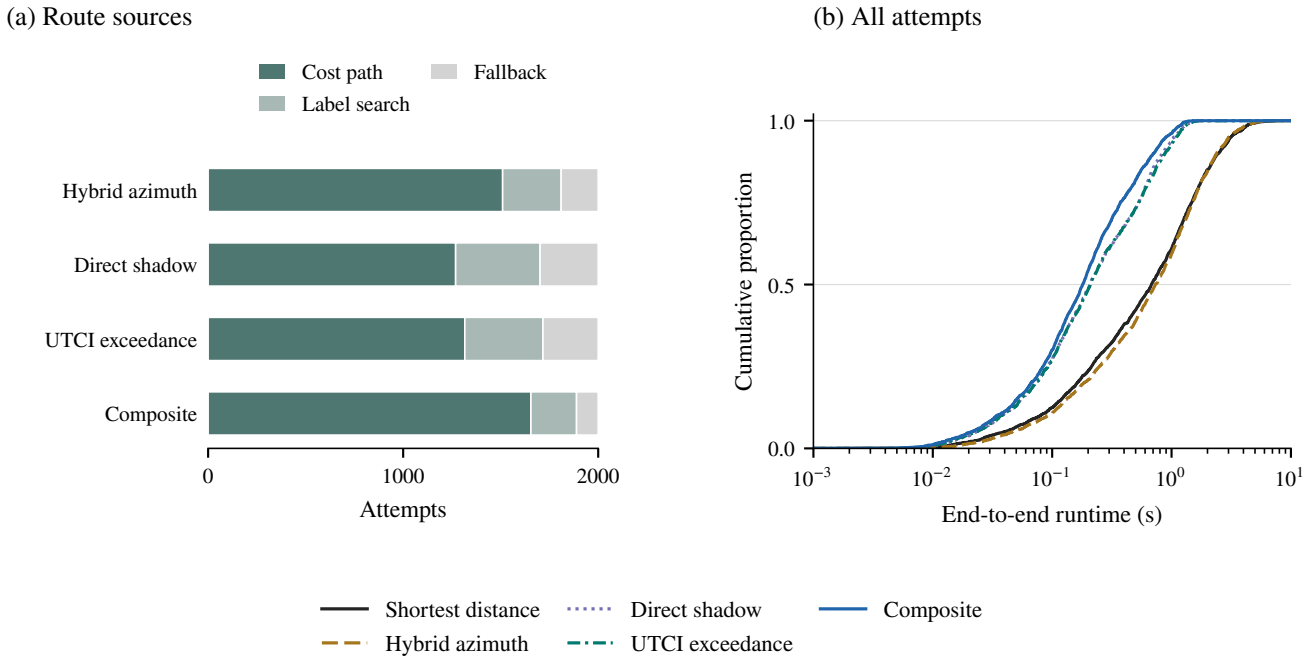


Figure 3: Returned-route sources and end-to-end runtime. (a) Environmental attempts split into the initial cost path, label search, and shortest-path fallback. (b) Empirical cumulative runtime distributions include all attempts for every method. The logarithmic time axis includes the full observed tails.

and feature extraction, with shared caches and a fixed method order within each shard. Cache state, concurrent jobs, and host differences prevent a controlled solver-speed ranking. The 1.5 s cap applies only to the label loop.

6 Discussion and Limitations

The experiment compares engineering objectives within one search implementation. The fixed cohort and explicit fallback records make their outcomes directly comparable, but do not compare independent routing algorithms. The exact checks expose cap-induced losses even on 33-node networks (Appendix A); no optimality gap is available for the DC requests. A city-scale comparison with an exact or tightly bounded constrained-path solver remains necessary before claiming search quality.

The spatial and temporal scope is one modeled DC hour. The study cannot establish transfer across cities, seasons, neighborhoods, or times of day, and the fixed field ignores changes during a walk. The OD sample targets covered network nodes rather than observed demand. It therefore cannot estimate population benefits or typical pedestrian trips.

Street centerlines may lie in sun while a sidewalk is shaded, or the reverse. Source layers have different vintages; the traffic join uses a nearest-midpoint proxy in geographic coordinates without a maximum distance. Canopy does not guarantee shade at the pedestrian's position. Hybrid shade mixes geometry with cool UTCI samples and lacks independent shadow validation.

The composite remains uncalibrated. Thermal terms overlap, scenic and traffic scores are proxies, and the uncertainty input is zero. The one-factor checks describe local sensitivity at one allowance; they do not identify optimal weights or cover interactions. The small differences between objective settings require denser rerouting before being treated as a stable ranking. Sampling checks within the same modeled field do not test its physical accuracy.

The study has no field transects, participant study, or health measurements. Mean exceedance and exceedance–distance can disagree, and neither establishes comfort, safety, or clinical benefit. Independent environmental measurements and sidewalk-level networks are needed to test whether the modeled route differences survive at pedestrian height and position.

Code and Data Availability

The public artifact snapshot includes code, the exact OD pairs, compressed route records, source/input checksums, analysis scripts, and figure inputs. Git LFS stores the versioned graph and rasters. The README gives rerun and analysis commands. Manifests preserve each shard's configuration and the routing source hashes; the original execution environment and thermal-field generation are only partly documented. The archived run and joint-weight exploration are retained separately and do not enter the revised result tables.

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References

[1] R. Basu, N. Colaninno, A. Alhassan, and A. Sevtsuk. 2024. Hot and bothered: Exploring the effect of heat on pedestrian route choice behavior and accessibility. *Cities* 155 (2024), 105435. doi:10.1016/j.cities.2024.105435

[2] G. Boeing. 2017. OSMnx: New methods for acquiring, constructing, analyzing, and visualizing complex street networks. *Computers, Environment and Urban Systems* 65 (2017), 126–139. doi:10.1016/j.compenvurbsys.2017.05.004

[3] P. Bröde, D. Fiala, K. Błażejczyk, I. Holmér, G. Jendritzky, B. Kampmann, B. Tinz, and G. Havenith. 2012. Deriving the operational procedure for the Universal Thermal Climate Index (UTCI). *International Journal of Biometeorology* 56, 3 (2012), 481–494. doi:10.1007/s00484-011-0454-1

[4] N. Colaninno, R. Basu, M. Hosseini, A. Alhassan, L. Liu, and A. Sevtsuk. 2025. A sidewalk-level urban heat risk assessment framework using pedestrian mobility and urban microclimate modeling. *Environment and Planning B: Urban Analytics and City Science* 52, 5 (2025), 1071–1090. doi:10.1177/23998083241280746

[5] L. Da, R. Chhibba, R. Jaiswal, A. Middel, and H. Wei. 2024. Shaded route planning using active segmentation and identification of satellite images. In *Proceedings of the 33rd ACM International Conference on Information and Knowledge Management*. 5205–5209. doi:10.1145/3627673.3679234

[6] DC Office of the Chief Technology Officer. 2020. Generalized Land Cover 2020. ArcGIS image service, Open Data DC. Created by PlanIT Geo for the District Department of Transportation Urban Forestry Division; accessed 17 July 2026. https://imagery.dcgis.dc.gov/dcgis/rest/services/UFD/Generalized_Land_Cover_2020/ImageServer

[7] DC Office of the Chief Technology Officer. 2020. Urban Tree Canopy 2020. ArcGIS image service, Open Data DC. Accessed 17 July 2026. https://imagery.dcgis.dc.gov/dcgis/rest/services/UFD/Urban_Tree_Canopy_2020/ImageServer

[8] DC Office of the Chief Technology Officer. 2023. Waterbodies—2023. Retrieved 28 October 2026; geographic display context only. https://maps2.dcgis.dc.gov/dcgis/rest/services/DCGIS_DATA/Planimetrics_2023/MapServer/14

[9] District Department of Transportation. 2023. Traffic Volume—2023. ArcGIS feature layer, Open Data DC. Accessed 17 July 2026. https://opendata.dc.gov/datasets/DCGIS::2023-traffic-volume/about

[10] District Department of Transportation. 2026. Roadway Block: Geographic Context Snapshot. DC GIS Roadway Block service, layer 163; retrieved 28 October 2026; CC BY 4.0; display context only. https://opendata.dc.gov/pages/roadway-centerlines

[11] District of Columbia. [n. d.]. Sustainable DC 2.0: Transportation. Historical commute mode share for 2016; accessed 28 October 2026. https://sustainable.dc.gov/sites/default/files/dc/sites/sustainable/page_content/attachments/SDC2%20Transportation.pdf

[12] European Environment Agency. [n. d.]. Thermal Comfort Indices—Universal Thermal Climate Index, 1979–2020. Archived indicator documentation; standard UTCI stress categories; accessed 28 October 2026. https://climate-adapt.eea.europa.eu/en/metadata/indicators/thermal-comfort-indices-universal-thermal-climate-index-1979-2019

[13] A. A. Hagberg, D. A. Schult, and P. J. Swart. 2008. Exploring network structure, dynamics, and function using NetworkX. In *Proceedings of the 7th Python in Science Conference*. 11–15. doi:10.25080/TCWV9851

[14] Stefan Irnich and Guy Desaulniers. 2005. Shortest Path Problems with Resource Constraints. In *Column Generation*, Guy Desaulniers, Jacques Desrosiers, and Marius M. Solomon (Eds.). Springer, 33–65. doi:10.1007/0-387-25486-2_2

[15] H. G. Kamath, N. Sudharsan, M. Singh, N. Wallenberg, F. Lindberg, and D. Niyogi. 2026. SOLWEIG-GPU: GPU-Accelerated Thermal Comfort Modeling Framework for Urban Digital Twins. *Journal of Open Source Software* 11, 118 (2026), 9535. doi:10.21105/joss.09535

[16] N. Kolaxidis, C. Ludwig, S. Knoblauch, J. Fürle, K. Foshag, S. Fendrich, S. Lautenbach, and A. Zipf. 2025. Mitigating heat stress by reducing solar exposure in pedestrian routing. *Transactions in GIS* 29, 6 (2025), e70110. doi:10.1111/tgis.70110

[17] M. Levenson, D. Pearlmutter, and O. Aleksandrowicz. 2025. An observational analysis of shade-related pedestrian activity. *Buildings and Cities* 6, 1 (2025), 398–414. doi:10.5334/bc.574

[18] F. Lindberg, B. Holmer, and S. Thorsson. 2008. SOLWEIG 1.0: Modelling spatial variations of 3D radiant fluxes and mean radiant temperature in complex urban settings. *International Journal of Biometeorology* 52, 7 (2008), 697–713. doi:10.1007/s00484-008-0162-7

[19] NOAA Office for Coastal Management. 2022. 2022 DC OCTO Lidar DEM: Washington, DC. NOAA InPort item 69939. Accessed 17 July 2026. https://www.fisheries.noaa.gov/inport/item/69939

[20] NOAA Office for Coastal Management. 2022. 2022 DC OCTO Lidar: Washington, DC. NOAA InPort item 69938. Accessed 17 July 2026. https://www.fisheries.noaa.gov/inport/item/69938

[21] OpenStreetMap contributors. 2026. OpenStreetMap. Open database licensed under ODbL 1.0. Accessed 17 July 2026. https://www.openstreetmap.org/copyright

[22] A. Patel, N. Sudharsan, T. Brooks, H. Kamath, D. Crawley, M. Coudert, Z. Baumer, and D. Niyogi. 2026. CoolPath Tool: A Thermal Comfort Path Planning Tool for Urban Mobility. arXiv preprint. arXiv:2602.02540 doi:10.48550/arXiv.2602.02540

[23] Georgios Skoumas, Klaus Arthur Schmid, Gregor Jossé, Matthias Schubert, Mario A. Nascimento, Andreas Züfle, Matthias Renz, and Dieter Pfoser. 2015. Knowledge-Enriched Route Computation. In *Advances in Spatial and Temporal Databases (Lecture Notes in Computer Science, Vol. 9239)*. Springer, Cham, 157–176. doi:10.1007/978-3-319-22363-6_9

[24] A. Vartholomaios. 2023. Follow the shade: Detection of optimally shaded pedestrian paths within the historic city center of Thessaloniki. *IOP Conference Series: Earth and Environmental Science* 1196 (2023), 012070. doi:10.1088/1755-1315/1196/1/012070

[25] D. Verma, O. Mumm, and V. M. Carlow. 2026. CoolPaths: Street-scale Physiological Equivalent Temperature (PET) mapping and cooler-routes planning using open data. *City and Environment Interactions* 30 (2026), 100349. doi:10.1016/j.cacint.2026.100349

[26] H. Wolf, A. R. Vierø, and M. Szell. 2025. CoolWalks for active mobility in urban street networks. *Scientific Reports* 15 (2025), 14911. doi:10.1038/s41598-025-97200-2

[27] L. Yang, T. Miyata, S. Kuroki, T. Urakubo, and P. Finnerty. 2025. Route planning in urban environments: A shade-aware navigation framework. In *2025 IEEE 14th Global Conference on Consumer Electronics (GCCE)*. 644–645. doi:10.1109/GCCE65946.2025.11275334

[28] P. Zippenfenig. 2023. Open-Meteo.com Weather API. Computer software, Zenodo. doi:10.5281/zenodo.7970649

A Exact Search Check

We generate 100 graphs in each of three families with seed 42: 17-node layered directed graphs (five layers of width three), nine-node cyclic directed graphs, and 11-node chains with two parallel edges per step. Edge lengths are 40–180 m; thermal and contextual features are generated numerical values. An independent depth-first enumerator retains every simple edge path, including parallel-edge choices. It shares the configured edge-cost function with the router, so this tests search rather than validating the cost model.

Three hundred graphs, four allowances, and UTCI/composite objectives give 2,400 reference cases. Of these, 1,395 have a constrained optimum strictly above the unconstrained optimum. With the wall-clock cap disabled, the default eight-label registry and 20,000-expansion limit match every reference within relative tolerance 10^{-8} and respect every distance ceiling. Removing the expansion cap with registry sizes one, eight, or 32 also matches all cases. A deliberately tight 20-expansion condition matches 1,687 of 2,400 cases; its capped fallbacks can be suboptimal. Frozen generated graphs, raw results, and the enumerator are shared. Agreement on these small graphs supplies no optimality bound for the DC network.

To activate the actual default cap, we also generate 25 chains with 32 binary parallel-edge choices: 33 nodes and 2^{32} possible paths each. Short edges have integer lengths of 40–70 m and exceedance of 8–12 K; longer edges add 10–50 m with exceedance of 0–4 K. An independent dynamic program retains minimum cost at every integer distance. Across 200 graph–objective–allowance cases, the default count limits match 50 exact references and truncate 150 searches. Mean relative cost gap is 42.3%, with a maximum of 136.0%. These deliberately difficult cases diagnose cap behavior; they do not estimate error on the DC network. Both checks disable the wall-clock cap.

B Engineering Feature Definitions

Let $\mathrm{clip}(x) = \min(1, \max(0, x))$. Canopy and land cover are sampled on the densified centerline; building height is additionally sampled at lateral offsets of one-half and one estimated street width.

Let c_e be the fraction of canopy samples at least 2 m high; k_e the mean canopy height divided by 12 m and clipped; v_e the larger of canopy- and land-cover-derived greenery fractions; and w_e the water fraction. The scenic proxy is

$$g_e = \mathrm{clip}(0.60c_e + 0.25k_e + 0.10v_e + 0.05w_e). \tag{9}$$

For nearest-midpoint AADT a_e and the traffic layer's maximum $a_{\max}$, the traffic proxy is $n_e = \log_{10}(a_e + 1)/\log_{10}(a_{\max} + 1)$. This is not a decibel scale.

For solar altitude α, azimuth θ, bearing β_e, and estimated street width W_e, define

$$q_e(z) = \mathrm{clip}\left(\frac{z|\sin(\theta - \beta_e)|}{W_e \max(\tan\alpha, 10^{-6})}\right). \tag{10}$$

If explicit width is missing, $W_e = \max(4, 3.2p_e)$ m for p_e lanes, or 10 m if lane count is also missing. With mean and maximum canopy/building heights, use $z_e^c = \max(\bar{z}_e^c, 0.5z_e^{c,\max})$ and $z_e^b = \max(\bar{z}_e^b, 0.6z_e^{b,\max})$. The geometry shade proxy is

$$s_e^{\mathrm{geom}} = \mathrm{clip}\Big(\max\{c_e, q_e(z_e^b), c_e q_e(z_e^c)\}\Big), \tag{11}$$

set to one below the horizon. For local fractional hour t,

$$\tau_c(t) = [25.5 + 3.2\cos(2\pi(t - 14)/24)]^\circ\mathrm{C}, \tag{12}$$

$$s_e^{\mathrm{cool}} = (n_e^u)^{-1} \sum_i \mathbf{1}[u_{ei} \le \tau_c(t)], \tag{13}$$

$$s_e^{\mathrm{az}} = \max(s_e^{\mathrm{cool}}, s_e^{\mathrm{geom}}). \tag{14}$$

At $t = 14$, the cool threshold is 28.7°C. These coefficients define a hybrid surrogate, not a fitted shadow model.

C Composite Cost

Let signed elevation change be Δz_e, grade $m_e = \Delta z_e/l_e$, and $r_e = 10|m_e| + [\Delta z_e]_+/(25\,\mathrm{m})$. Edge time is t_e, estimated as $l_e/(1.35\,\mathrm{m\,s^{-1}})$ when unavailable. With ϕ_e from Eq. 7, the default cost is

$$A_e = 0.15\frac{l_e}{1000\,\mathrm{m}} + 0.20\frac{t_e}{600\,\mathrm{s}}, \tag{15}$$

$$C_e = A_e + \frac{\max(l_e, 1\,\mathrm{m})}{100\,\mathrm{m}}\big[\phi_e + 0.45(1 - g_e)^2 + 0.45n_e^2 + 0.35r_e + 0.15\sigma_e\big], \tag{16}$$

$$C(P) = \sum_{e \in P} C_e. \tag{17}$$

All ratios are dimensionless. $\sigma_e = 0$ throughout this experiment. The one-factor checks change the corresponding coefficients and shade multiplier separately. The 1 m floor prevents a zero environmental scale on a malformed zero-length edge. These formulas specify the engineering score used in the corrected rerun.